

Derivative of the sigmoid function:

The sigmoid function is:

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

Derivative,

$$\sigma'(x) = \sigma(x) (1 - \sigma(x))$$

Derivation,

write the chain rule, here, $\sigma(x)$

$$\frac{d}{dx} (1+e^{-x}) = -e^{-x}:$$

$$\sigma'(x) = -1 \cdot (1+e^{-x})^{-2} \cdot (e^{-x}) = \frac{e^{-x}}{(1+e^{-x})^2}$$

Now write the numerator as $e^{-x} = (1+e^{-x}) - 1$:

$$\sigma'(x) = \frac{(1+e^{-x}) - 1}{(1+e^{-x})^2}$$

$$= \frac{1}{1+e^{-x}} - \frac{1}{(1+e^{-x})^2}$$

$$= \sigma(x) - \sigma(x)^2$$

$$\text{So, } \sigma'(x) = \sigma(x) (1 - \sigma(x))$$